ADIC MORPHISMS INTO A FORMAL COMPLETION COME FROM MORPHISMS MEETING THE CLOSED SUBSET ONLY AT THE CLOSED POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own misspelling 'incusion' for 'inclusion'; it lies inside the counted span and is reproduced verbatim, as the delivered text is the author's.

Editorial note. The proof displays one identification of fibre products; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms from the spectrum of a complete local Noetherian algebra into an algebraic space agree with morphisms from its formal spectrum. Same chapter, Lemma lemma-map-into-algebraic-space, invoked at proof lines 7327-7329. Produces the unique morphism into X lifting the composition with the comparison map.
- The formal completion along a closed subset has the expected type, in particular is adic*. Same chapter, Lemma lemma-formal-completion-types at proof lines 7323-7325, together with Definition definition-adic-morphism at proof lines 7325-7326. Makes the terminology of an adic morphism applicable and gives representability.
- The comparison map from the formal completion to the ambient space is a monomorphism. Same chapter, the monomorphism property of the comparison morphism, used at proof lines 7336-7339. Identifies the two fibre products over the completion and over the ambient space.
- Krull's intersection theorem. Standard, invoked at proof lines 7348-7350. Concludes that the two ideals defining the relevant fibre products coincide.

Source reference. Original lemma 0AQ1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4752–4752. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0AQH: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7227–7227. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over S. Assume X and Y are locally adic*. We say f is an *adic morphism* if f is representable by algebraic spaces. See discussion above.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $T \subset |X|$ be a closed subset such that $X \setminus T \to X$ is quasi-compact. Let R be a complete local Noetherian S-algebra. Then an adic morphism $p : Spf(R) \to X_{/T}$ corresponds to a unique morphism $g : \mathrm{Spec}(R) \to X$ such that $g^{-1}(T) = \{\mathfrak{m}_R\}$.*

Proof. The statement makes sense because $X_{/T}$ is adic* by Lemma 0AQ1 (and hence we're allowed to use the terminology adic for morphisms, see Definition 0.1). Let p be given. By Lemma 0AQH we get a unique morphism $g : \mathrm{Spec}(R) \to X$ corresponding to the composition $\mathrm{Spf}(R) \to X_{/T} \to X$. Let $Z \subset X$ be the reduced induced closed subspace structure on T. The incusion morphism $Z \to X$ corresponds to a morphism $Z \to X_{/T}$. Since p is adic it is representable by algebraic spaces and we find

$$\mathrm{Spf}(R) \times_{X_{/T}} Z = \mathrm{Spf}(R) \times_X Z$$

is an algebraic space endowed with a closed immersion to $\mathrm{Spf}(R)$. (Equality holds because $X_{/T} \to X$ is a monomorphism.) Thus this fibre product is equal to $\mathrm{Spec}(R/J)$ for some ideal $J \subset R$ which contains $\mathfrak{m}_R^{n_0}$ for some $n_0 \geq 1$. This implies that $\mathrm{Spec}(R) \times_X Z$ is a closed subscheme of $\mathrm{Spec}(R)$, say $\mathrm{Spec}(R) \times_X Z = \mathrm{Spec}(R/I)$, whose intersection with $\mathrm{Spec}(R/\mathfrak{m}_R^n)$ for $n \geq n_0$ is equal to $\mathrm{Spec}(R/J)$. In algebraic terms this says $I + \mathfrak{m}_R^n = J + \mathfrak{m}_R^n = J$ for all $n \geq n_0$. By Krull's intersection theorem this implies $I = J$ and we conclude. $\square$